

Notebook Calculus

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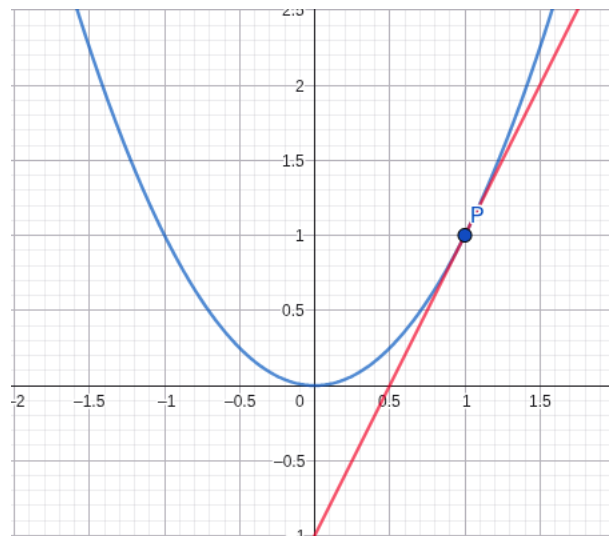
1 Limits and Derivatives

1.1 The Tangent and Velocity Problems

1.1.1 The Tangent Problem

EXAMPLE 1: Find a equation for the tangent line to the parabola $y = x^2$ at the point $P = (1, 1)$.

My solution:



For to find a equation for a line we need two points in the line. For the problem the only point of the line is P . My idea is evaluate a second point in the tangent line that is in the parabola. For a good precision will select the point $(0.1, f(0.1))$:

slop:

$$m = \frac{1 - f(0.1)}{1 - 0.1} = 1.9$$

coefficient b :

$$b = 1 - 1 \cdot 1.9 = 0.9$$

Therefore, the equation for the tangent line is approx:

$$g(x) = 1.9x - 0.9$$

1.1.2 The Velocity Problem

EXAMPLE 3: Suppose that a ball is dropped from the upper observation deck of the CN Tower in Toronto, 450 m above the ground. Find the velocity of the ball after 5 seconds.

My solution: